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Conservation History as Living Context

Week 2 Monday — Chapter 2

NRES 315: Human Dimensions of Fish and Wildlife Management

Last updated: 2020-08-28

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Today's question

How do earlier eras of fish and wildlife management still shape a decision made now?

History explains the **institutions, funding, laws, professional habits, and public expectations** a manager inherits.

Source: Chapter 2, §§2.1–2.2.

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What you should be able to do

By the end of today, you should be able to:

- identify the defining priority and social context of each historical era;
- explain how an earlier era left a legacy for a current decision; and
- use a human-dimensions question to examine that legacy rather than treating it as neutral or inevitable.

Source: Chapter 2, guiding questions, §2.2.

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History asks more than "what happened?"

Priority What problem seemed most urgent?	Context Which values, economics, and politics shaped the response?	Legacy What remains in current management?
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Conservation history traces changing relationships among people, species, habitats, and institutions.

Source: Chapter 2, §§2.1–2.2.

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Conservation history is cumulative

Source: Chapter 2, Figures 2.1 and §§2.1.1–2.2.6.

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A timeline tells order; history explains inheritance

Timeline question What came next?	Historical question What did this era leave behind for today's decisions?
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An inherited legacy might be a legal framework, a source of agency revenue, an idea about expertise, or a pattern of inclusion and exclusion.

Source: Chapter 2, §2.2.

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Indigenous stewardship is the first layer
The history of North American fish and
wildlife management begins long before
European colonization.



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Indigenous stewardship: reciprocal responsibility

Many Indigenous nations developed place-based stewardship systems over millennia.

- Humans are part of relationships with living and nonliving beings.
- Stewardship can be guided by respect, relevance, responsibility, and reciprocity.
- Knowledge is created and shared across generations through oral and visual traditions.

Source: Chapter 2, §2.2.1.

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"Pristine wilderness" is an incomplete history

Framing conservation history as beginning with Muir or Roosevelt overlooks the active management of pre-colonial landscapes.

Landscapes were actively shaped by Indigenous peoples: Karuk and Yurok controlled fire; Lakota and Blackfeet developed bison-harvest practices; Lummi and Yakama created salmon fisheries.

Ignoring that stewardship erases Indigenous cultures and can help justify removal from ancestral lands.

Source: Chapter 2, §2.2.1.

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The legacy continues through governance

Contemporary collaboration and co-management include:

- Mille Lacs Lake fishery management and treaty-rights partners;
- Menominee forest management and a seventh-generation responsibility;
- Indigenous input to the Alexander Archipelago wolf assessment; and
- Hawaiian marine governance that integrates customary values.

Source: Chapter 2, §2.2.1.

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Abundance became a utilitarian premise
1600–1850: European settlers commonly perceived North American resources as effectively endless.

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Era of Abundance: freedom, opportunity, and commodities

European restrictions on hunting had often served monarchs and aristocrats. In North America, fewer restrictions helped make wildlife seem like:

- a freely usable resource;
- a symbol of freedom and opportunity; and
- a source of profit in market hunting, trapping, and the fur trade.

Perceived abundance helped normalize a belief that use did not require long-term restraint.

Source: Chapter 2, §2.2.2.

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Era of Exploitation: short-term use, lasting damage

Conditions	Consequences
• few regulatory frameworks; • industrial-scale markets; • new harvest technologies; and • westward expansion.	• depleted deer and beaver; • reduced predator ranges; • declining salmon; and • near extermination of bison.

Source: Chapter 2, §2.2.3 and Figure 2.2

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Bison shows why a social history matters

The near extermination of bison reflected a convergence of:

- industrial hide harvest and market demand;
- military and government campaigns against Indigenous peoples; and
- westward expansion.

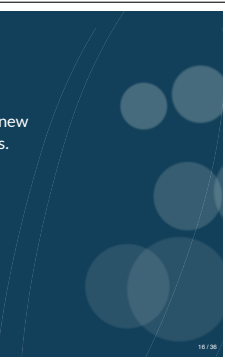
Ecological decline is often inseparable from political power, economic systems, and human relationships.

Source: Chapter 2, §2.2.3

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Protection changed the rules
1900–1930: overharvest and species decline produced organized advocacy, new laws, and new conservation institutions.



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Era of Protection: from commodity to public concern

The passenger pigeon and Carolina parakeet were extinct; bison had fallen to roughly a thousand animals by the 1890s.

- Harriet Hemenway and Minna Hall organized opposition to the plume trade and helped establish the Massachusetts Audubon Society.
- The Lacey Act (1900) restricted interstate commerce in illegally taken wildlife.

Source: Chapter 2, §2.2.4 and Figure 2.3

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Protection required cooperation beyond one state

The Migratory Bird Treaty Act (1918):

- followed a U.S.–Great Britain treaty acting on Canada's behalf;
- made most migratory birds illegal to hunt, capture, or sell; and
- recognized that migratory species require management at a continental scale.

Source: Chapter 2, §2.3.4

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Game management built a profession
1930–1960: management moved from
reactive protection toward proactive,
science-based restoration and sustainable
use.

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Era of Game Management: a durable funding legacy

Program	Revenue source	What it supported
Pittman-Robertson Act (1937)	Excise tax on sporting arms and ammunition	State wildlife restoration, habitat, research, public hunting areas
Dingell-Johnson Act (1958)	Excise tax on certain fishing tackle	Sport-fish restoration and related state projects

The model linked purchases by hunters, anglers, and later boaters to public conservation benefits.

Source: Chapter 2, §2.2.5.

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Expertise, education, and organizations expanded

Professionalization created a durable infrastructure:

- Leopold's *American Game Policy* (1930) and *Game Management* (1933);
- a Wisconsin wildlife degree program and Cooperative Wildlife Research Units;
- The Wildlife Society and Ducks Unlimited (both 1937); and
- a land ethic that framed conservation as a moral duty of care.

Source: Chapter 2, §2.2.5.

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A successful model can still be narrow

The game-management era elevated game and sportfish, but it also created costs:

- nongame species and broader ecological problems received less priority;
- public outreach often remained one-way education; and
- programs could displace Indigenous communities, overlook knowledge, and reproduce racial and economic exclusions.

Source: Chapter 2, §2.2.3.

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Environmental management broadened the frame
1960–1990: conservation increasingly
addressed ecosystems, environmental
harm, public participation, and
accountability.

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Era of Environmental Management: systems, not isolated species

Rachel Carson's *Silent Spring* (1962) helped connect pesticide use to ecological and human-health consequences.

The management question widened: How do habitats, ecosystems, pollution, and human systems interact?

The social question widened too: Who participates, and how are institutions accountable?

Source: Chapter 2, §2.2.4.

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Law made participation and accountability more visible

National Environmental Policy Act (1970)

Institutionalized public participation and accountability in environmental decision-making.

Endangered Species Act (1973)

Made protection of imperiled species a national conservation priority.

Source: Chapter 2, §2.2.6.

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Human dimensions became a named professional field

Early information from hunters often served "surrogate biology": estimating harvest and population measures.

The emerging field added different questions: landowner-hunter conflict, recreation experiences, wildlife values, economic effects, and public participation.

In 1973, a North American Wildlife and Natural Resources Conference session formally introduced "human dimensions in wildlife programs."

Source: Chapter 2, §2.2.6, Boxes 2.1-2.2.

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Institutionalization changed who agencies could learn from

Development	What it made more possible
Missouri hired Dan Witter (1978)	Full-time agency human-dimensions expertise
Cornell's Human Dimensions Research Unit	Dedicated university research capacity
Yale's Human Dimensions of Wildlife Study Group	Professional scholarly community
Later journals and university programs	A recognized field and shared methods

Source: Chapter 2, §2.2.6, Boxes 2.1-2.2.

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Globalization and international governance change the scale
1990–present: conservation problems and solutions increasingly cross ecological, political, and social boundaries.

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Global challenges require connected governance

Contemporary conservation faces interconnected challenges:

- biodiversity loss and landscape fragmentation;
- wildlife trade and invasive species;
- climate change and shifting species distributions; and
- urbanization and changing relationships with wildlife.

When a problem crosses boundaries, management needs cooperation among institutions, communities, and publics at multiple scales.

Source: Chapter 2, §2.2.7.

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International frameworks meet local knowledge

Global agreements and local governance operate together:

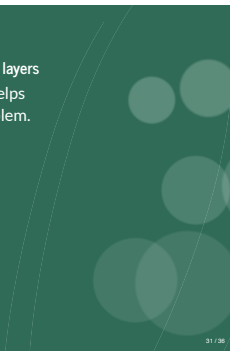
- CBD and CITES help align national and local work with global priorities.
- The European Union's Common Fisheries Policy uses public engagement and adaptive management.
- Namibia's community conservancies connect local knowledge, livelihoods, wildlife coexistence, and conservation goals.

Source: Chapter 2, §2.2.7.

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Modern challenges reveal the accumulated layers
The history becomes useful when it helps
diagnose a present conservation problem.



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A present decision is social, ecological, and institutional

For urban wildlife, predator reintroduction, habitat restoration, or climate adaptation, a manager may need to ask:

- Which publics experience benefits, costs, or risk?
- Which authorities and knowledge systems have a role?
- What history shapes funding, trust, access, or participation?
- What new challenges make older approaches insufficient?

Source: Chapter 2, §2.2.8.

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Technology adds capacity—and new questions

New possibilities GIS, remote sensing, data science, iNaturalist, eBird, and social media can expand research and participation.	New responsibilities Managers must consider access, data quality, privacy, misinformation, and who benefits from technology.
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Source: Chapter 2, §2.2.8.

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Practice: trace one legacy forward

With a partner, choose one Chapter 2 era. Prepare a 45-second explanation:

1. What priority, institution, or assumption is associated with that era?
2. How could it still shape a current fish or wildlife decision?
3. What human-dimensions question would a manager ask before treating that legacy as neutral or inevitable?

Source: Chapter 2, §2.2.

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From history to public responsibility

History shows that management inherits layered priorities and institutions.

Wednesday asks the next question:

For whom should those institutions act, and how are they accountable?

Source: Chapter 2, §§2.2–2.3; Chapter 3, §§3.1–3.3.

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Exit check

Complete both statements:

1. One current management decision is shaped by the legacy of _____ because _____.
2. A human-dimensions question about that legacy would ask _____.

Source: Chapter 2, §§2.2–2.3.

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